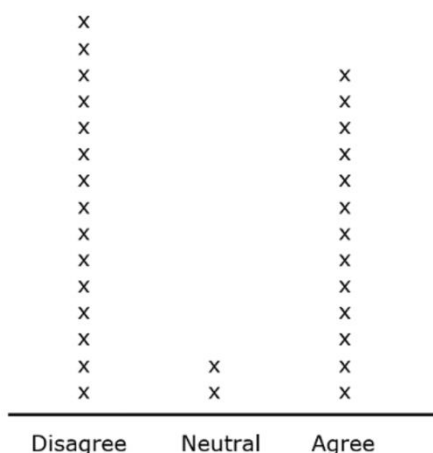


Grade 6 Accelerated
Unit 10
Lesson 9 Homework

Name Answer key
Date _____
Period _____

A middle school in Warner Robins, GA is considering requiring students to wear uniforms for the upcoming school year. The decision was based on proposals by the school advisory committee and PTSA. Parents, teachers, and staff overwhelmingly agreed that wearing uniforms will have a positive effect on the school. The school's principal conducted a random survey of 30 students to gauge student opinions on the potential mandatory uniform policy. The results are shown on the line plot below. Use the data to answer questions 1-5.



1. What fraction of students surveyed were neutral on the potential mandatory school uniform policy?

$$\frac{2}{30} \approx 6.67\%$$

2. A study conducted by a research group in Nevada indicated that 54% of middle school students agreed with a mandatory uniform policy. Explain how this compares to the students at the school in Warner Robins that agree with a mandatory uniform policy.

The 54% is higher than the 43% sample at the school in Warner Robins.

3. There are approximately 3,512 middle students in Warner Robins, Georgia public schools. Predict the number of students in Warner Robins that would disagree with a mandatory uniform policy based on the data from the line plot.

50% of 3,512 = 0.5 (3,512) = 1,756 students would disagree.

4. There are approximately 3,512 middle students in Warner Robins, Georgia public schools. Predict the number of students in Warner Robins that would be neutral about a mandatory uniform policy based on the data from the line plot.

6.67% of 3,512 = .0667 (3,512) $\approx$ 234 students would be neutral.

5. Explain at least *two* factors that could contribute to the differences between your estimates and the actual number of students who could have voted about a mandatory uniform policy.

Answers may vary.

- 1) A sample of 30 students is very small for a population of 3512 students.
- 2) The sample did not indicate how the random survey was given, which could create a biased sample.

A guidance counselor at Top Performance High School in Washington, DC wants to analyze scores on the SAT to increase the average scores for seniors. The counselor gathered the SAT scores for all seniors at the school. The average SAT score in the US is 1051.

The counselor picked a random sample of 200 senior scores and recorded the information in the table below. Use this data to answer questions 6-10.

Students that scored under 1051	Students that scored 1051 and higher.
86	114

6. What is the proportion of students that scored 1051 or greater on the SAT at Top Performance High School?

$$\frac{114}{200}$$

7. Estimate the probability of a senior in the random sample scoring 1051 or greater on the SAT? Explain your method.

$$\frac{114}{200} = 0.57 = 57\% \quad \text{divide number of students scoring 1051 or greater by total in sample.}$$

8. There are 638 seniors at Top Performance High School. Using equivalent fractions, how many seniors are expected to score 1051 or greater on the SAT (based on the sample data)?

$$\frac{114}{200} \times \frac{3.19}{3.19} \approx \frac{364}{638} \quad 364 \text{ students}$$

9. If you use the percent method to answer question 8, what similarities and differences do you notice?

$$0.57 (638) \approx 364 \quad \text{Similarity: same answer with different method.}$$

Difference: multiply percent by 364.

10. The guidance counselor estimates that 400 seniors will score 1051 or greater on the SAT sample data sample data. Explain whether this would be considered an unreasonable based on the sample data.

400 students are considerably more than the 364 estimated to score 1051 or greater, so this could be considered unreasonable.